

Causal Inference & ML

TA Session 1

Data Science for Decision Making

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Outline

- Panel Datasets
- Panel Data Models
- Fixed Effects vs Random Effects
- Hausman Test

Corollary

For any gamble in , there is some probability such that.

Example 1

- bla bla bla
- bla2 bla2 $\propto$ bla2

Python

```
1 import pandas as pd
2 import numpy as np
3 import linearmodels as lm
4
5 data = pd.DataFrame({
6     'entity': np.repeat(np.arange(1, 6), 4),
7     'time': list(np.arange(2000, 2004)) * 5,
8     'y': np.random.randn(20),
9     'x1': np.random.randn(20),
10    'x2': np.random.randn(20)
11 })
12 data.head(5)
```

	entity	time	y	x1	x2
0	1	2000	0.485317	1.025849	2.202466
1	1	2001	-0.667812	-1.032976	-0.420256
2	1	2002	1.125400	-0.719898	-2.206085
3	1	2003	0.101090	-1.671555	1.398242
4	2	2000	1.059610	0.642749	0.521411